General Relativity Final Examination: Marking Summary

Each agent's `answers.tex` in `runs/` was marked against `solutions.tex`. There are 20 marks in total: Q1 (2), Q2 (3; a: 2, b: 1), Q3 (4; a: 2, b: 2), Q4 (5; a: 2, b: 1, c: 2), Q5 (6; a: 1, b: 2, c: 3). Questions are numbered 1–5 as they appear in the paper.

Notes on the reference solutions. `solutions.tex` gives the de Sitter scale factor as $a(t) = L^2\cosh^2(t/L)$. That is a^2; the scale factor is $a(t) = L\cosh(t/L)$. Every agent gave $L\cosh(t/L)$, and this was marked correct. The reference also writes $\hat{\Gamma}_{[\nu\rho]}{}^{\mu} = \Gamma_{\nu\rho}{}^{\mu}$ in Q2b where it means the symmetric part, $\hat{\Gamma}_{(\nu\rho)}{}^{\mu}$. Agents were marked on the physics, not on these slips.

Model	Effort	Q1 /2	Q2 /3	Q3 /4	Q4 /5	Q5 /6	**Total /20**	Comments
claude-haiku-5-5	low	2	3	4	5	6	**20**	All parts correct, with full working. Answers are mislabelled "Question 2"–"Question 6" (the point values were read as question numbers); no marks deducted. Q3b has a garbled remark ("r is not a good coordinate … but v is"; it is t that fails), but the main argument is right.
claude-haiku-5-5	high	2	3	4	5	6	**20**	All parts correct. Derives the Kretschmann scalar from orthonormal-frame Riemann components. Checks both Friedmann equations. Gives the interferometer phase shift $\Delta\phi = 4\pi Lh/\lambda$.
claude-sonnet-5-5	low	2	3	4	4.5	6	**19.5**	Concise and correct throughout, apart from Q4c. The derivation and the square Penrose diagram are correct, but the shaded region labelled "visible to an observer at $\chi = 0$" (the triangle above $\eta = 0$ next to the pole) is the causal future of a point on the worldline, not the observer's causal past. It also contradicts the caption, which correctly says the visible region is bounded by the anti-diagonal (-0.5).
claude-sonnet-5-5	high	2	3	4	5	6	**20**	All parts correct and thorough. Proves the Lorenz gauge is attainable. Shows the light cones tipping inside the horizon in EF coordinates. Gives the proper time πGM to the singularity. Gives the chirp scalings $f \propto (t_c - t)^{-3/8}$.
claude-opus-5-5	low	2	3	4	5	6	**20**	All parts correct. Includes a Friedmann check, horizon analysis on the Penrose diagram, and the chirp mass. One minor slip in non-required material: $K(r{=}2GM)$ is quoted as $3/(4G^2M^2)$ instead of $3/(4G^4M^4)$. No marks deducted.
claude-opus-5-5	high	2	3	4	5	6	**20**	All parts correct; the most complete script. Shows the result is independent of the connection-index convention and links it to the Levi-Civita uniqueness theorem. Includes infall proper time, tidal-force estimates, and the inverse EF metric. The Penrose diagram labels the static patch and the event and particle horizons. Discusses the quadrupole power, $\dot{f}$, the chirp mass, antenna patterns and the standard siren.

Overall. Five of the six scripts scored full marks; claude-sonnet-5-5 (low) lost half a mark. The scripts differ mainly in depth and supporting material, not correctness. The high-effort Opus and Haiku answers were the most detailed; the low-effort Sonnet answer was the shortest. All six runs compiled `answers.pdf`. Several left LaTeX auxiliary files behind because the sandbox's permission checks blocked deletion. This is not assessed in the marks.